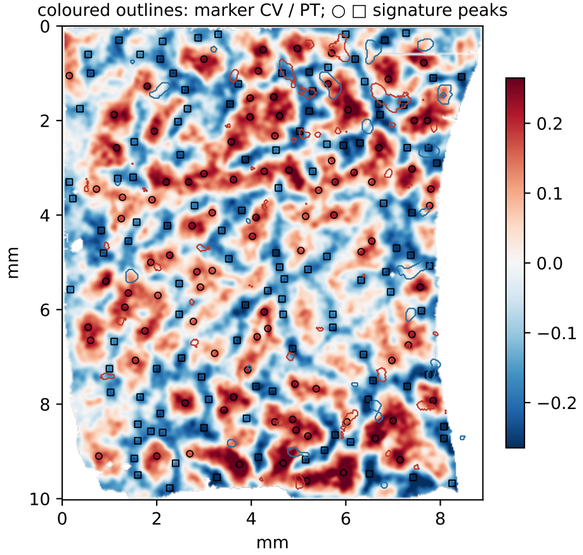
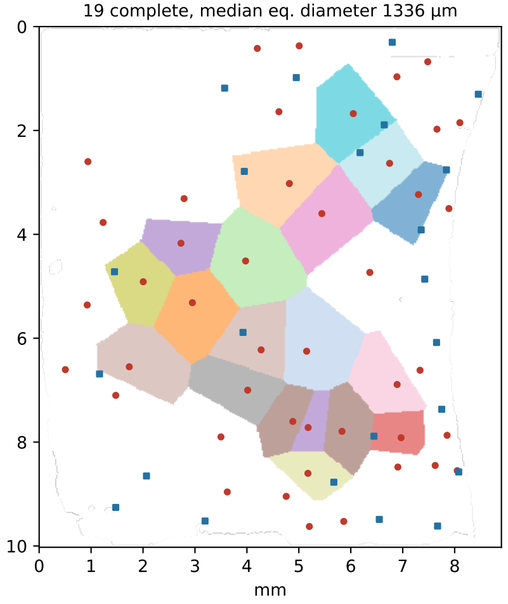
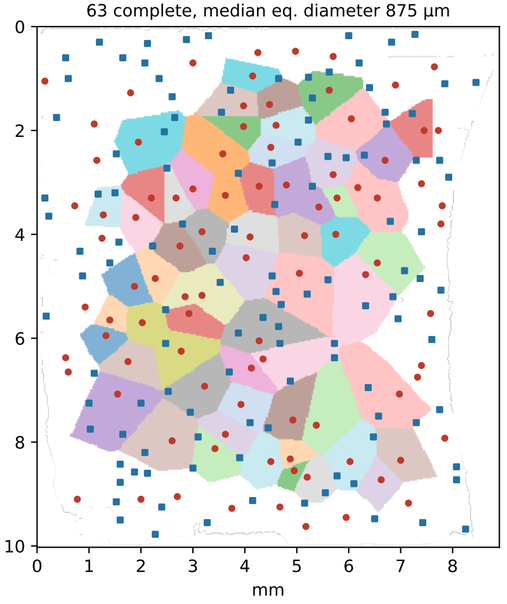
A Zonation signature (red = pericentral)
coloured outlines: marker CV / PT; ○ □ signature peaks



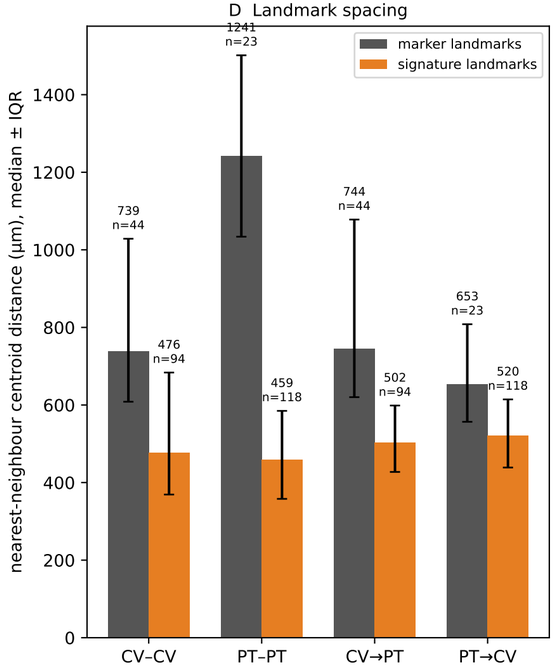
B Voronoi lobules, marker landmarks
19 complete, median eq. diameter 1336 µm



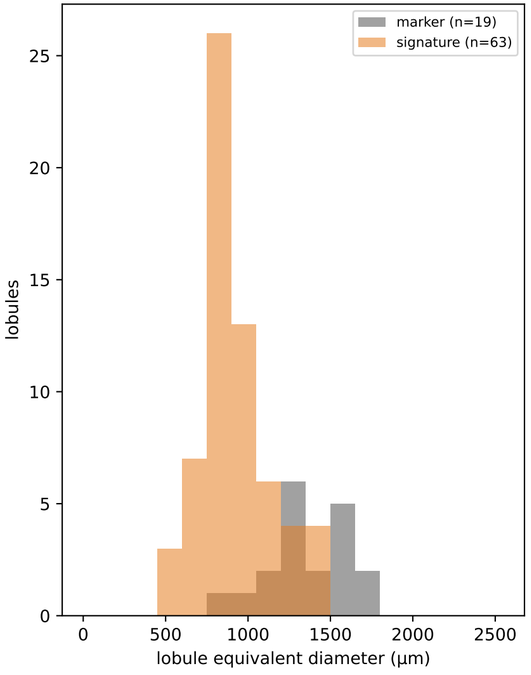
C Voronoi lobules, signature landmarks
63 complete, median eq. diameter 875 µm



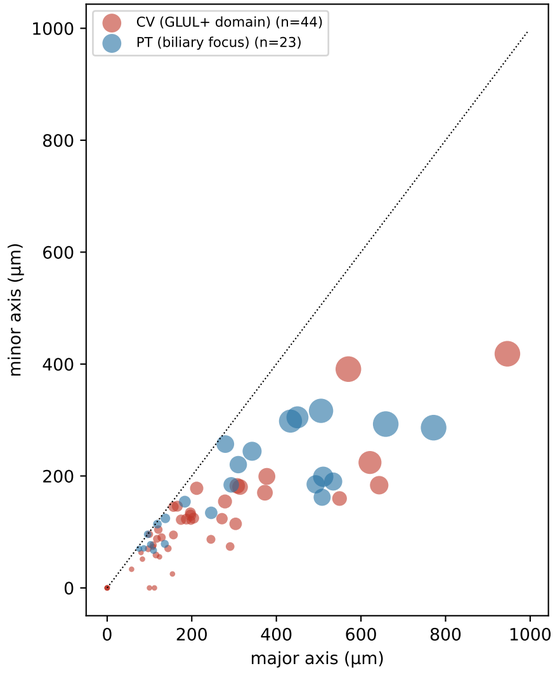
D Landmark spacing



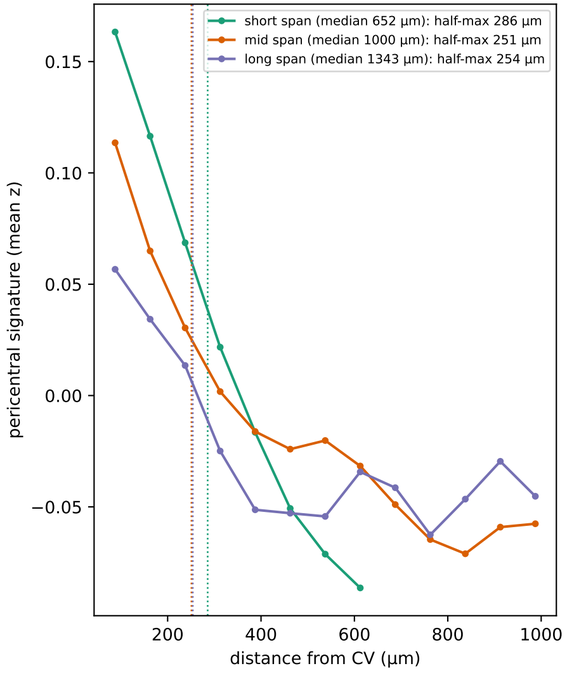
E Lobule size (CV-centred Voronoi, complete cells)



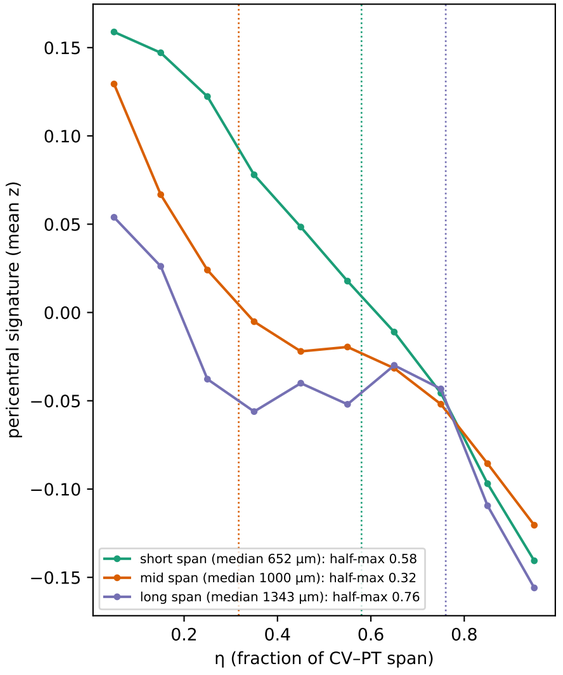
F Marker landmark size and elongation
(GLUL+ domain = pericentral hepatocyte cuff, not lumen)



G Pericentral score vs absolute distance



H Pericentral score vs relative position η



Spread of half-max position across span tertiles (coefficient of variation, block-bootstrap median)
marker PTs: µm 0.12 vs η 0.35 (η more invariant in 13% of resamples); marker + PP-signature PTs: µm 0.42 vs η 0.28 (η more invariant in 80%); half-max µm by tertile: 283, 446, 125